1) I wrote a program that takes a text file full of binary digits and removes everything that isn't a 0 or 1. It then groups the bits into either 7-bit or 8-bit chunks and converts them into ASCII characters. If a chunk equals 8 (backspace), it deletes the last character. My program tries both 7-bit and 8-bit decoding and chooses the one that produces the most readable output.

   When I ran it on *infection.txt*, the binary decoded into a readable message. This shows that my decoder works because it can turn a long binary string into human-readable ASCII text. The infection file was simply a binary-encoded message, and my program successfully decoded it.